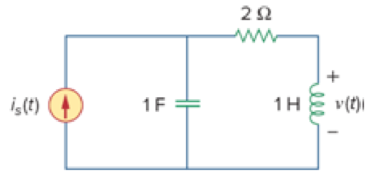
Chapter 18, Problem 60P

Problem

A band-limited signal has the following Fourier series representation: $is(t) = 10 + 8\cos(2\pi t + 30°) + 5\cos(4\pi t - 150°)$mA

If the signal is applied to the circuit in Fig. 18.55, find $v(t)$.

Figure 18.55



Step-by-step solution

Step 1 of 4

Refer to Figure 18.55 in the textbook.

Draw the circuit as shown in Figure 1.

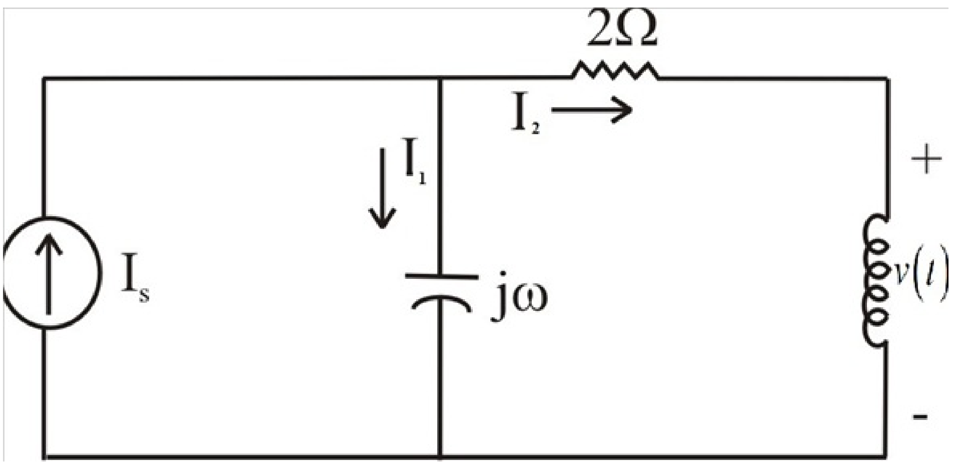


Figure 1

Step 2 of 4

Calculate the current I_2.

Apply current division rule.

$$I_2 = I_s\left(\frac{\left(\frac{1}{j\omega}\right)}{2 + j\omega + \left(\frac{1}{j\omega}\right)}\right)$$

$$= I_s\left(\frac{1}{2j\omega + (j\omega)^2 + 1}\right)$$

$$= I_s\left(\frac{1}{2j\omega - \omega^2 + 1}\right)$$

Step 3 of 4

Determine the voltage across inductor $v(t)$.

$$v = j\omega I_s\left(\frac{1}{2j\omega-\omega^2+1}\right) \quad \text{...... (1)}$$

Consider $i_s(t)=10$.

For dc current, the voltage across inductor is zero.

Therefore, the voltage across inductor for dc current is zero.

Consider $i_s(t)=8\cos(2\pi t+30°)$.

Replace 2π for ω in equation 1.

$$\begin{aligned}
v &= \frac{8(2\pi\angle 90°)}{1-(2\pi)^2+j2(2\pi)} \\
&= \frac{8(2\pi\angle 90°)}{1-4\pi^2+j4\pi} \\
&= \frac{50.27\angle 90^o}{-38.48+j12.566} \\
&= 1.2418\angle -71.92^o \text{ V}
\end{aligned}$$

Step 4 of 4

Consider $i_s(t)=5\cos(4\pi t-150°)$.

Replace 4π for ω in equation 1.

$$\begin{aligned}
v &= \frac{5(4\pi\angle 90°)}{1-(4\pi)^2+j2(4\pi)} \\
&= \frac{5(4\pi\angle 90°)}{1-16\pi^2+j8\pi} \\
&= \frac{62.83\angle 90^o}{-156.91+j25.13} \\
&= 0.3954\angle -80.9^o \text{ V}
\end{aligned}$$

Write the voltage across inductor as,

$$\begin{aligned}
v(t) &= 0+1.2418\cos(2\pi t-71.92°+30°)+0.3954\cos(4\pi t-80.9-150°) \\
&= 1.2418\cos(2\pi t-41.92°)+0.3954\cos(4\pi t-230.9°) \\
&= 1.2418\cos(2\pi t-41.92°)+0.3954\cos(4\pi t+129.1°) \text{ mV}
\end{aligned}$$

Therefore, the voltage across inductor is

$\boxed{1.2418\cos(2\pi t-41.92°)+0.3954\cos(4\pi t+129.1°)\text{ mV}}$.